Early Interplanetary Smallsat and Microsat Studies at JPL: 1978-1994

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This poster provides a summary of the earliest spacecraft- and mission-related studies conducted at JPL focused on applying then-emerging small and 'micro' satellite technologies to various interplanetary and deep-space mission objectives, including science missions and technology-demonstration missions.

Of course, many of the first satellites developed and launched at JPL (including the U.S.'s very first launched satellite, *Explorer 1,* launched in 1958) rightly fit today's definition for small satellites. The typical size of JPL-designed interplanetary spacecraft grew over subsequent decades, leading to very large craft such as *Galileo* and *Cassini*. This poster thus captures the history of the 'second wave' of small JPL interplanetary spacecraft concepts, which started in 1978 and continued until NASA's "Faster, Better, Cheaper" era began in the early 1990s.

The term "microspacecraft" in this context first appeared in 1981 as the title of a JPL advanced interplanetary spacecraft concept report. The overall concept of microspacecraft for solar system exploration became embedded in the JPL culture in 1988 following a seminal paper authored by one of the co-authors of this summary (Jones), who also organized a broad and pathfinding workshop on the subject that summer.

Mission concepts based on small- and micro-sized spacecraft proliferated at JPL in the late 1980s and early 1990s, leading to several mature mission and system concept studies for missions to the Moon, near-Earth asteroids, Mars and even Pluto. Several novel technologies and mission-design techniques resulted from this flurry of creative work, and several of these advancements favorably impacted subsequent NASA missions.

This poster highlights the genesis of this work at JPL and traces the many interesting mission and system concepts, advanced technology developments, programmatic milestones and mission applications that occurred during this 15+-year period, including some key non-JPL developments for context.

Late 1970s

Studies of Shuttle-based interplanetary smallsat concepts (Staehle, et al.)

1978 | 1979

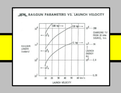
"There are a great many discoveries waiting to be made in the Solar System. Right now we have available the tools to make these discoveries. Small-scale planetary missions can augment our rapidly developing capabilities for large-multidiscipline missions."

-- Rob Staehle, *Small Planetary Missions for the Space Shuttle*, 1979

Early 1980s

First system-level design concepts for "microsats" (J. Burke, et al.)

Studies of railgun-launched microsats (R. Jones, et al.)



1980 | 1981 | 1982 | 1983 | 1984

"While the laws of physics do impose some fundamental size constraints, they do not forbid the building of deep-space craft significantly smaller than our present ones."

-- Jim Burke, *Micro-Spacecraft*, 1981

Late 1980s

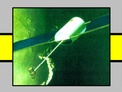
Studies of railgun-launched and rocket-launched microsats (R. Jones, et al.)

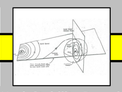
← Assessments of SDIO-related miniaturized technologies →

← Pegasus-class missions →

1985 | 1986 | 1987 | 1988 | 1989

Lunar Get-Away Special (*LGAS*) spacecraft and mission design (K. Nock, et al.)



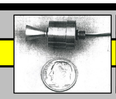


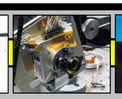
'Weak Stability Boundary' trajectories discovered

Workshop on Microspacecraft for Space Science

Early 1990s

Mission studies of fast sprints to Pluto using microspacecraft; evolved into *Pluto Fast Flyby* mission concept (Staehle, t al.)



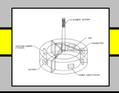


Extensive evaluation and testing of miniaturized microsat components

Clementine launched

NEAR and *Mars Pathfinder* approved

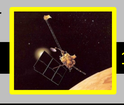
Lunar Observer pre-project subsatellite concept



Workshop on Microtechnologies for Space Systems

"Faster, Better, Cheaper" kicks in

Mars Observer failure



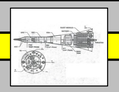
New Millennium Program (tech-demo)

NMP absorbs MDP

Deep Space 1

1990 | 1991 | 1992 | 1993 | 1994

CRAF penetrator design



Lunar missions from Pegasus winged rocket

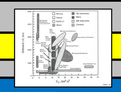
Weak Stability Boundary trajectory applied to *Hiten* mission

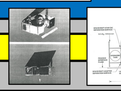
Microspacecraft Devlopment Program

Dan Goldin as NASA Administrator →

Other deep-space smallsat and microsat mission studies

MSTI to Mars study





JPL reorganizes for small/moderate missions

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